Optoelectronic lab

Experiment 5: Laser diodes

1 Basics

1.1 Principle

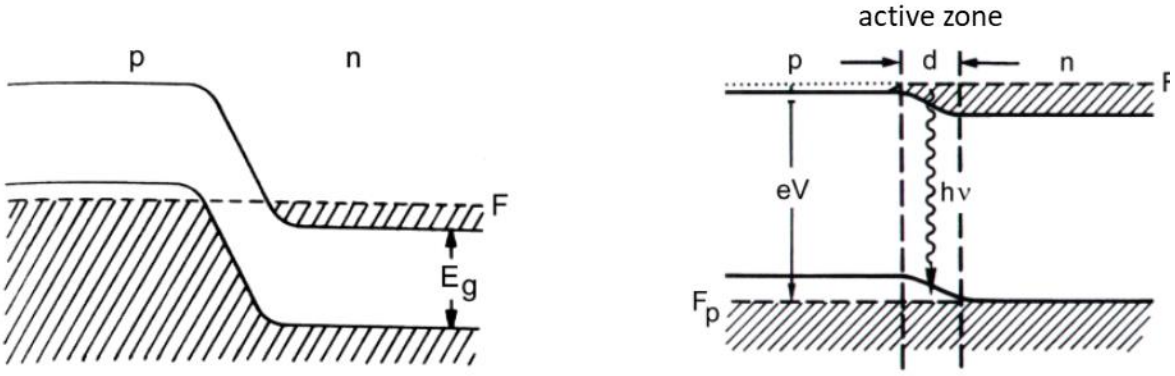


***Fig. 1**: Band diagram of a highly doped pn junction in a voltage-free state (left) and with applied forward voltage (right)*

Most lasers utilize transitions between discrete molecular or atomic energy levels. In semiconductor lasers, however, the laser transition takes place between the conduction and valence band, i.e. between quasi-continuous states. The first laser condition is fulfilled if a population inversion is produced, i.e. electrons are lifted into the conduction band by a pumping process, which relax very quickly to the band edge and from there can recombine with holes in the valence band. This pumping process can take place optically, i.e. by excitation with light, or electrically, i.e. by the injection of minority charge carriers. As with the LED, a p-n junction is required for this.

In this case, we speak of a laser diode. By switching the p-n junction in the forward direction, a population inversion can occur at the junction so that stimulated emission is possible. In the simplest case, the interfaces of the semiconductor serve as resonator mirrors, which reflect a considerable proportion of the radiation back into the semiconductor due to the usually high refractive index of the semiconductor material used (n ≈ 3.5). This is referred to as a *Fabry-Perot resonator*.

The following applies to the reflection coefficient for vertical irradiation (see also the description of the LED experiment)

$$R = \left(\frac{n_{HL} - n_{Luft}}{n_{HL} + n_{Luft}} \right)^2 \qquad (1)$$

- How high is the reflection for the GaAs (n=3.5) - air (n=1) interface?

The reflection is nevertheless much lower than on conventional laser mirrors. As the resonator losses are therefore high, the pumping must be correspondingly strong; this means that the laser threshold is only exceeded at relatively high current densities.

1.2 Structure

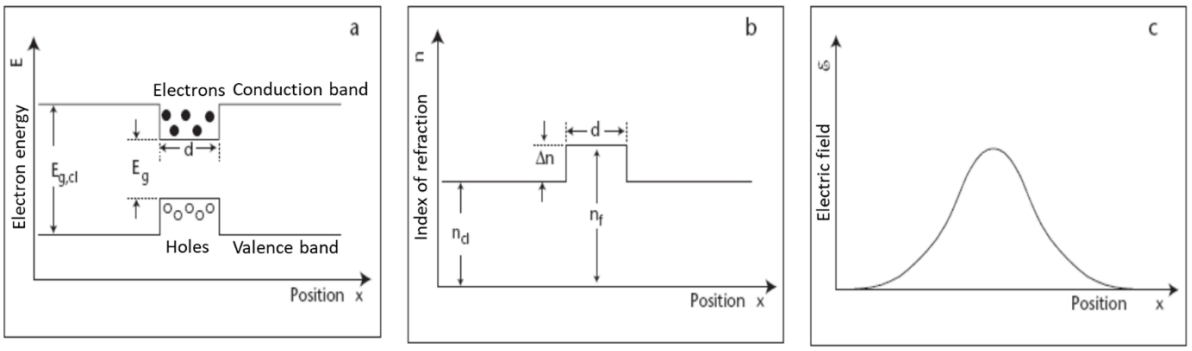


***Fig. 2**: Confinement of electrons, holes and electric field by a double heterostructure in vertical direction x of a laser diode: (a) energy diagram with conduction and valence band, (b) profile of the refractive index and (c) field distribution of the optical fundamental wave, which is guided in the double heterostructure. (Source: Ferdinand-Braun-Institut für Höchstfrequenztechnik, Berlin)*

As with LEDs (see LEDs experiment), double heterostructures are used for laser diodes, in which a semiconductor with a lower band gap is embedded in a semiconductor with a higher band gap. This concentrates the recombination in a small area.

In addition, the semiconductor with the smaller band gap usually has a higher refractive index. This means that the active region is also an optical waveguide, so that the photons are also concentrated in the active region, which significantly reduces the laser threshold currents. A measure of the proportion of light power contained in the active region is the confinement factor Γ (ideally $\Gamma = 1$).

In so-called broad area lasers, there is only vertical confinement; the active area extends over the full width of the laser or at least over a large area (approx. 100 µm). However, many semiconductor lasers use strip waveguides (Fig. 3), which have a confinement in two dimensions due to the lateral structuring of the semiconductor. Another advantage is the limitation of the current path. It is even sufficient to structure only the top layer, as this also laterally modifies the effective refractive index in the active area (ridge waveguide laser).

Most laser diodes are now based on quantum films. In these, the active area is reduced even further and the laser becomes even more efficient (also due to the changed density of states in quantized structures). However, such a quantum film alone can no longer be used as a waveguide, as the thickness is only a fraction of the wavelength. In this case, "Separate Confinement Heterostructure (SCH)" structures are used, in which the quantum film is placed centrally in a waveguide.

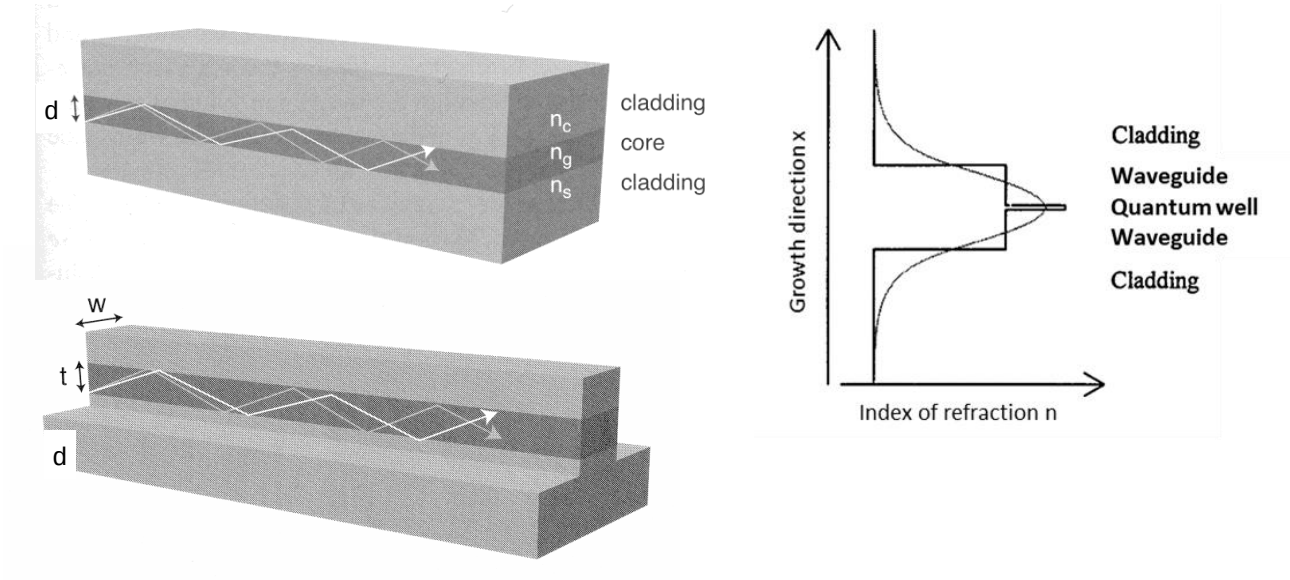


***Fig. 3**: Left: Laser diode without and with lateral waveguide. From [3]. Right: Separate-confinement heterostructure. The active area is the quantum film; the wave is guided in the waveguide between the two so-called cladding layers.*

1.3 Threshold current density

1.3.1 Connection with laser setup

The laser threshold results from the condition that the gain in the laser-active medium must compensate for the losses when the light circulates, the *threshold gain* is thus given by

$$\gamma_{th} = \alpha_0 + \frac{1}{2L}\ln\left(\frac{1}{R_1 R_2}\right) \tag{2}$$

This generally applies to all lasers. For the laser diode, we now assume the same reflection coefficients for both interfaces $R = R_{12} = R$. We also replace γ_{th} with $\Gamma\gamma_{th}$, where $\Gamma < 1$ is the confinement factor (see (section 1.2)). We thus have

$$\Gamma\gamma_{th} = \alpha_0 + \frac{1}{L}\ln\left(\frac{1}{R}\right) \text{ or } \gamma_{th} = \frac{1}{\Gamma}\left(\alpha_0 + \frac{1}{L}\ln\left(\frac{1}{R}\right)\right) \tag{3}$$

On the other hand, the gain is related to the population inversion (N_2 -N_1) and to the recombination rate of the charge carriers $1/\tau$ (τ lifetime of the charge carriers) (not derived here):

$$\gamma = \frac{\lambda^2}{8\pi\tau}(N_2 - N_1)g(\nu) \tag{4}$$

Here, $g(\nu)$ is the line shape function of the laser-active transition.

In order to move on to measurable variables, we estimate the number of charge carriers recombining per unit of time via the injected current (more precisely: via the current density):

$$\gamma = \frac{(N_2 - N_1)}{\tau} \approx \frac{j}{qd} \tag{5}$$

Here j is the current density, d the thickness of the active region and q the charge of a charge carrier; the expression on the right-hand side thus describes the number of charge carriers injected into the active region per unit time.

Inserting into (4) gives

$$\gamma = \frac{\lambda^2}{8\pi}\left(\frac{j}{qd}\right)g(\nu) \tag{6}$$

and in combination with (3) we obtain the *threshold current density*:

$$j_{th} = \frac{8\pi q}{\lambda^2 g(\nu)}\frac{d}{\Gamma}\left(\alpha_0 + \frac{1}{L}\ln\left(\frac{1}{R}\right)\right) \tag{7}$$

This somewhat confusing expression is nevertheless valuable for understanding if you take from it what promotes low threshold current densities:

- The first fraction contains quantities that are important for all lasers, e.g. it is generally the case that the laser threshold is easier to overcome for long wavelengths
- A thin active region is beneficial, hence the preferred use of quantum films in lasers, among other things
- A high confinement factor Γ is advantageous.
- Long resonators and high reflectivities are also favorable, and this also applies to all lasers in general. However, "long" is relative for laser diodes: the minimum length is around 100 µm.

The *threshold current* I_{th} is usually the variable to be measured directly; it is simply calculated by multiplying j_{th} by the area of the active zone.

1.3.2 Temperature dependence

As the temperature rises, non-radiative processes such as the scattering of charge carriers on phonons (lattice vibrations) generally increase. As a result, the threshold current density also increases with temperature.

This is determined by the empirical formula

$$I_{th}(T) = I_{th,0}\exp(T/T_0) \tag{8}$$

is described. The unit of T and T_0 is °C. T_0 usually has values between 50 °C and 200 °C and has no direct physical equivalent, but is a measure of the temperature sensitivity of the threshold current; the higher T_0, the lower the temperature sensitivity. $I_{th,0}$ stands for the threshold current at T=0°C.

1.4 PI curve

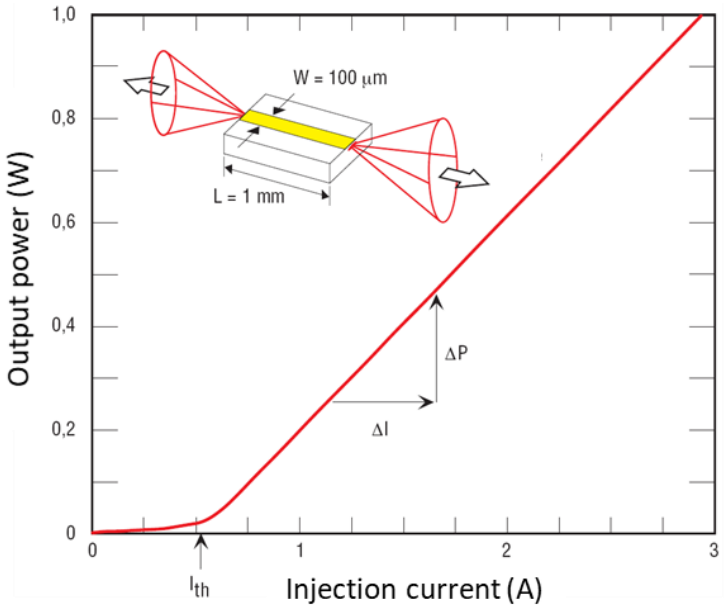


***Fig. 4**: P-I behavior of a laser diode (Source: Spectra Physics)*

Now that the threshold current density has been discussed, let's take a look at the decoupled power as a function of the injection current, i.e. the so-called P-I curve. The following applies to laser operation (i.e. for $I>I_{th}$ with I_{th} as the threshold current):

$$P = \frac{1}{2}\eta_i\eta_e\left(\frac{h\nu}{q}(I - I_{th})\right) \qquad (9)$$

- P is the decoupled power, the factor ½ is based on the fact that power is only decoupled at one of two (identical) mirrors.
- η_i stands for the internal quantum efficiency (i.e. the number of photons generated per injected charge carrier). This value should be close to 1.
- η_e stands for the external quantum efficiency (i.e. the number of decoupled photons per generated photon), this value is <1, as a large proportion of the photons are reflected back into the semiconductor and maintain the laser process. The following applies $\eta_e = \left(\frac{\alpha_S}{\alpha_S + \alpha_0}\right)$, where α_0 stands for the losses in the laser medium (equation (2)) and α_s =1/L ln (*1/R*) for the mirror "losses" due to back reflection (which, however, as already mentioned, maintain the laser process)
- the expression $h\nu/q$ describes the emitted power per injected charge carrier

The equation describes a straight line for $I>I_{th}$. Above the laser threshold, the following applies

$$\frac{dP}{dI} = \frac{\eta_i\eta_e h\nu}{2q} \qquad , (10)$$

Any increase in the injected current thus leads linearly to an increase in the coupled-out power, as is to be expected for a process based on stimulated emission. The "differential quantum efficiency" $\eta_d = \eta_i\eta_e$ is often used, which can be determined directly from the slope according to (10) and stands for the number of coupled-out photons per coupled-in charge carrier, and is therefore also a clear and very important parameter. This usually decreases with increasing temperature, again due to non-radiative processes (scattering by phonons).

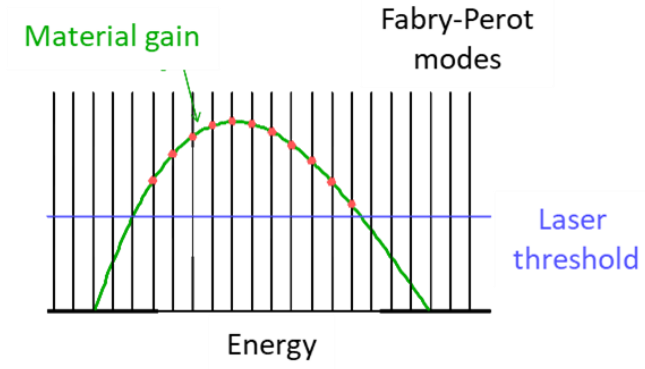
1.5 Emission

The emission wavelength is essentially determined by the laser-active transition, i.e. by the position of the gain profile. The main factor here is of course the band gap, which can be varied over a wide range in compound semiconductors. In the case of quantum film or quantum dot lasers, the quantization energy also plays a role.

The mirrors formed by the interfaces do not exhibit any pronounced wavelength dependence, so the laser usually oscillates at the energy at which the effective optical gain is maximum or the losses are minimum. The following results for the distance between the longitudinal modes (see also the DFB laser experiment description)

$$\Delta\nu = \frac{c}{2nL} \tag{11}$$

Where n is the refractive index of the active region and L is the length of the laser. As only the modes that can exceed the laser threshold oscillate, the laser emission is significantly narrower than the (spontaneous) emission below the laser threshold.



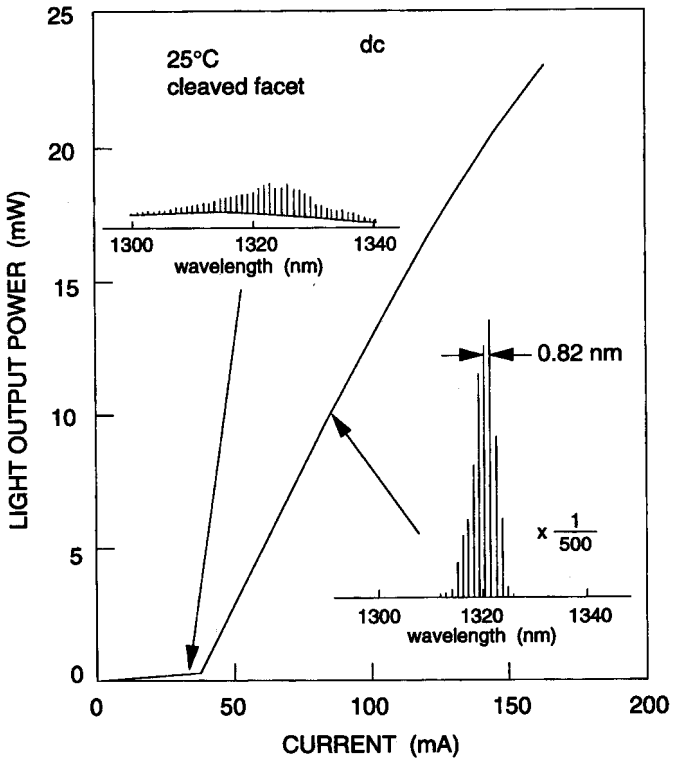


***Fig. 5**: Top: Gain profile and Fabry-Perot modes of a laser diode at the laser threshold. Right: Emission spectrum below and above the laser threshold. From [4]*

1.6 Fine tuning of the wavelength

Fine tuning of the wavelength can be achieved by changing the temperature or changing the current density.

- In semiconductors, an increase in temperature initially causes a reduction in the band gap so that the amplification profile shifts to longer wavelengths. In addition, the resonator expands and the refractive index usually increases so that the modes according to $\lambda_m = \frac{2nL}{m}$ also shift towards the long wavelengths. This effect is dominant,

so the mode spectrum shifts below the gain profile. An unpleasant side effect is the so-called mode jumps that can occur.

Precise adjustment of the laser temperature with the aid of temperature-controlled housings and temperature controls is therefore a great advantage, and efficient cooling is usually even necessary for powerful laser diodes.

- An increase in the operating current can lead to an increase in temperature and thus in turn to a change in the emission spectrum. As the converted power is quadratic in the current intensity, this dependency also applies to the shift in wavelength.

  In addition, the change in charge carrier density in the active zone also causes a variation in the refractive index and thus also shifts the mode spectrum of the resonator.

  Tuning via the current is quite critical, as the diode can be damaged or destroyed if the current is too high.

1.7 Radiation characteristics

In contrast to gas or solid-state lasers, semiconductor lasers have a strongly divergent emission. For this reason, laser diodes are usually operated in conjunction with collimator optics that parallelize the divergent beam.

To do this, however, you need to know the radiation characteristics of the laser diode. A distinction is made between near and far field.

1.7.1 Near field

Access to the near field is gained via a microscopic image of the exit surface. The near field is imaged directly onto a camera (CCD); if the exit surface is illuminated, orientation on the exit surface is easier.

1.7.2 Far field

When the light wave exits the resonator, diffraction effects are of great importance, causing the beam to widen. At some distance from the facet - approximately from a distance of w/λ^2_0 (w: width of the facet) - we speak of the far field of the laser diode. However, the theoretical description of light propagation inside and outside the resonator is not entirely straightforward. Mathematically, the far field results from the near field via a Fourier transformation.

In the far field, the light propagates in a light cone whose opening angle Θ depends on the thickness of the waveguide d and the wavelength λ_0 in free space:

$$\Theta = \lambda_0 / \pi d \qquad (12)$$

As a typical waveguide has a thickness of around 1 μm in the vertical direction, but is between 5 μm and 200 μm wide in the lateral direction, the laser beam of a diode is not round. It expands significantly more in the vertical direction (around 90°) than in the lateral direction (expansion of around 15°).

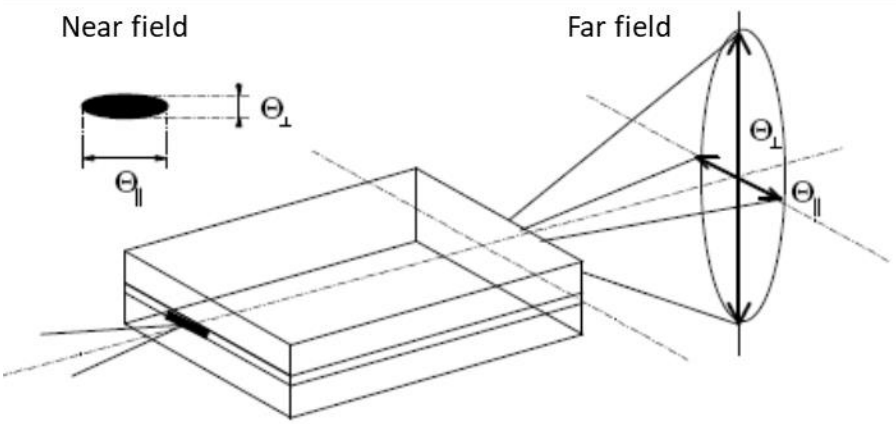


***Fig. 6**: Near and far field of a laser diode with Fabry-Perot resonator*

1.7.3 Measurement setup for near and far field

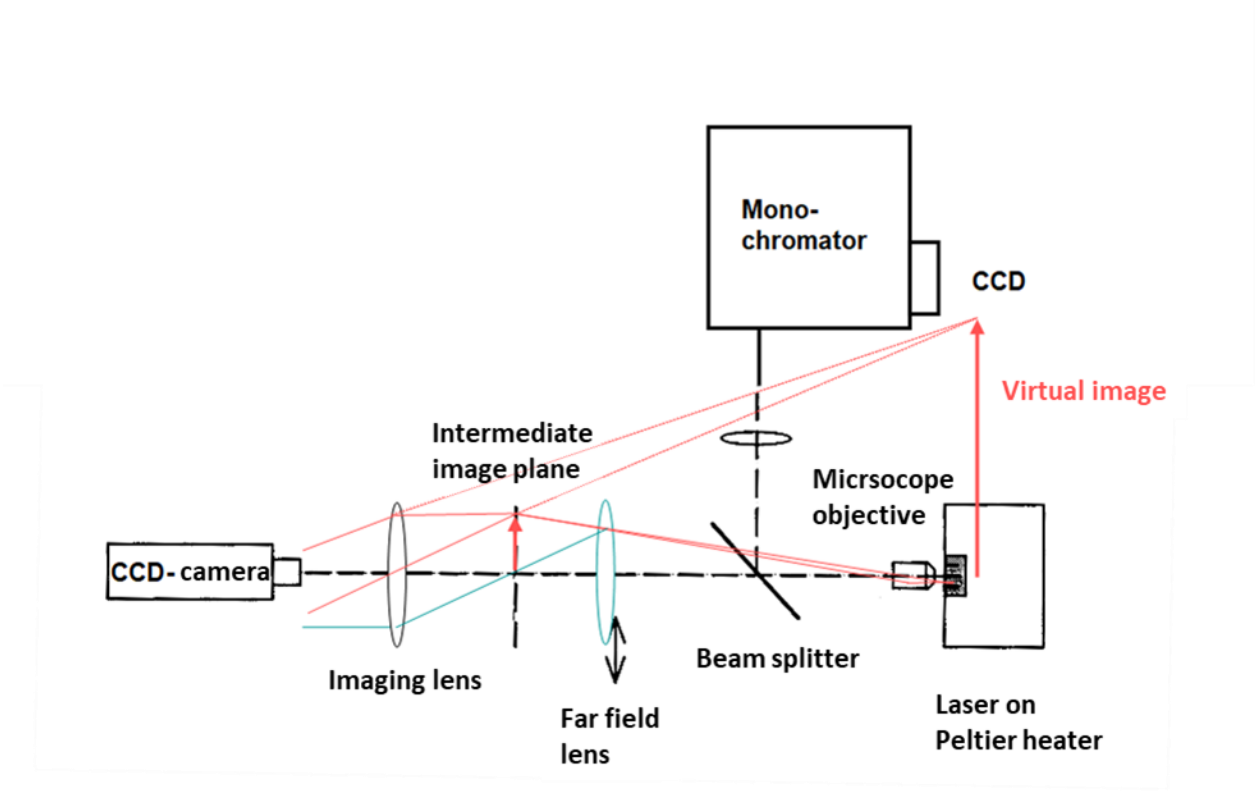


***Fig. 7**: Near and far field measuring station*

Near and far fields can be examined using the measurement setup (Fig. 7). The microscope objective and imaging lens represent a microscope (with an intermediate image plane drawn in), which is recorded with the aid of a camera.

Textbooks on optics are recommended for repeating the optical path in a microscope [5]. Here only in brief: If the far-field lens is not in place (red beam path), the object is magnified by the microscope objective and imaged onto the intermediate image plane. This real intermediate image is viewed through the eyepiece (imaging lens) as if through a magnifying glass, resulting in a further magnified virtual image (dashed red lines), which is recorded with the eye or (in this case) with the camera.

For far-field measurements, an additional lens (shown in blue) is inserted, which focuses the incident light rays onto the intermediate image plane (blue beam path). This causes the rays in front of the camera to be parallelized again by the imaging lens. This is tantamount to moving the virtual image to infinity. This means that the far field can be recorded directly with the camera.

2 Carrying out the experiment

2.1 Structure of the laser diode

The output power of laser diodes is extremely high compared to their dimensions. Laser diodes with a total volume of less than one cm^3 can emit several W of light output. However, this also means that operation under incorrect parameters can very quickly lead to the destruction of the diode.

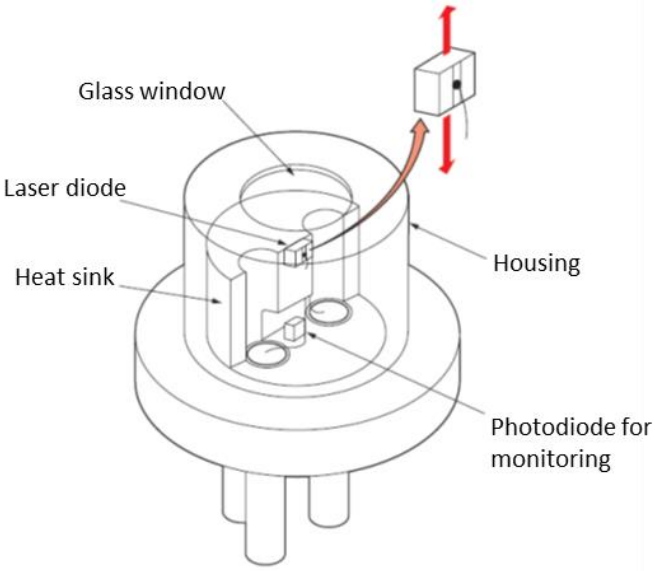


***Fig. 8**: Structure of a commercial laser diode*

Fig. 8 shows a schematic diagram of a housed laser diode. The actual diode is mounted on a heat sink (metal); some housings also contain Peltier cooling elements for temperature stabilization. The monitor photodiode allows power-regulated control.

These laser diodes can be plugged directly onto a (cooled if required) holder and controlled via regulated current sources (laser diode controller). It is possible to control both the current and the laser power. The first mode is used exclusively in the experiment; **it requires great care to prevent the diode from overheating**.

A laser diode with emission at λ=660 nm (red) is available for the experiment. It is mounted in a temperature-controlled holder, the temperature can be set between approx. 15 °C and 40 °C.

The maximum laser power is 2.5 mW. This radiation can already cause damage to the retina. Never look into the beam! If possible, always use the supplied attenuators (gray filters) in the beam before it hits the detectors.

2.2 Temperature control

The temperature control is already connected to the diode holder and set to a temperature of approx. 20 °C. To change the temperature, set the desired temperature using the software installed on the measuring computer. A small difficulty here: The temperature cannot be set directly, instead you must set the desired resistance of the temperature sensor in kΩ . The temperature values are assigned using the following calibration curve. After setting the resistance value, wait until the current temperature has stabilized at the set value.

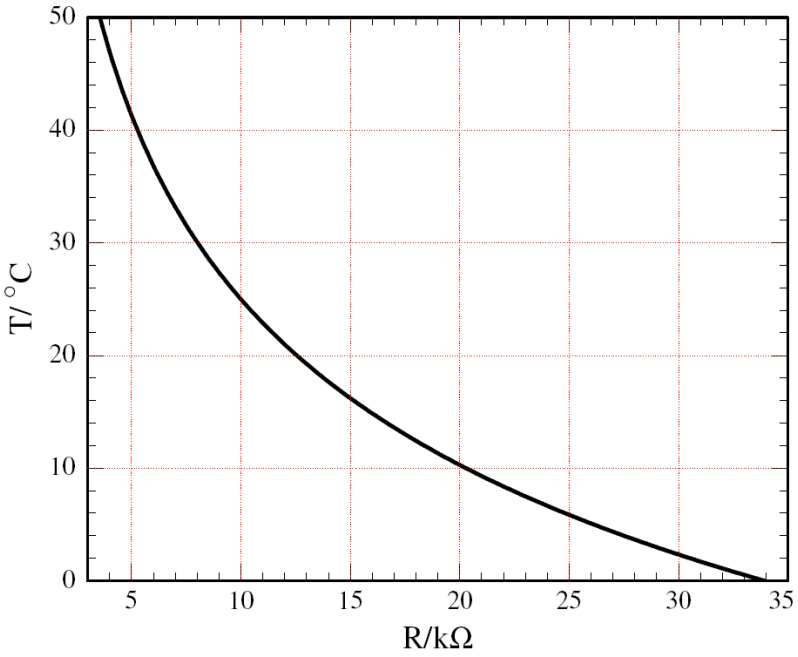


Fig. 9: *Calibration of the temperature control*

2.3 Laser diode control

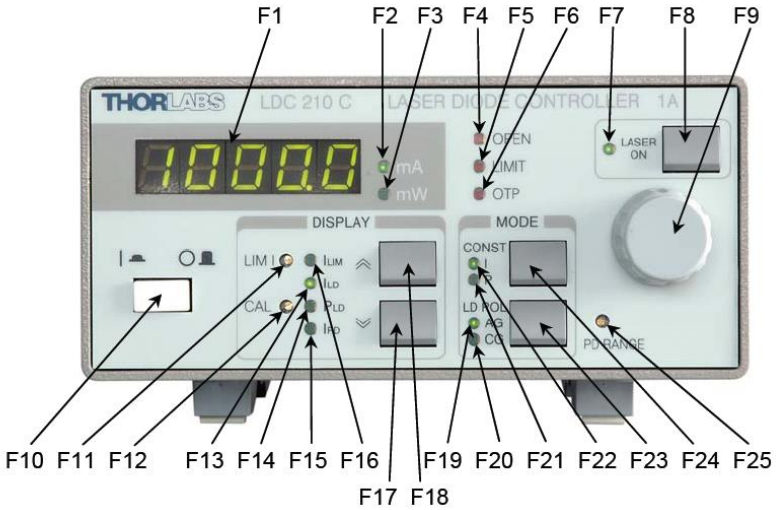


Fig. 10: *Front of the laser diode controller*

The diode control is also already connected to the diode holder. Only the constant current mode is used in the experiment:

- Turn controller F9 all the way to the left.
- Select Constant Current Mode by pressing button F24 until diode F22 lights up
- Press button F17 or F18 until diode I_{LD} (F13) lights up
- Switch the diode on via F8
- Set the current via F9. In principle, a current limiter prevents the diode from burning out, but be EXTREMELY careful here. Observe both the display F1 (laser operation is to be expected from approx. 20 mA) and the emission of the laser diode directly in the assembly.
- Press F17 or F18 to display either the fed-in current or the decoupled laser power.

2.4 Spectrometer

The spectra are recorded using a fiber optic spectrometer. Here, the light signal hits a fiber and is guided via it to a grating, spectrally split and detected with a CCD chip. The spectrometer is controlled entirely via the computer to which it is connected via USB.

The supervisor will give you a brief introduction to the software. **Protect the spectrometer from excessive intensities! Otherwise the CCD chip may be damaged. The maximum of the spectrum must not exceed the maximum count number for a longer period of time. If necessary, shorten the recording time or place gray filters in the beam path!**

3 Tasks

- Familiarize yourself with the entire setup (laser diode, control units, spectrometer, camera).
- Record the P-I characteristic curves for T=20°C, T=30°C and T=40°C. Determine the threshold current I_{th} and the gradient dP/dI for the respective temperatures and from this the differential quantum efficiency. Determine the values for T_0 and $I_{th,0}$ from your measurements (see Eq.8).
- Record spectra below (e.g. I=15 mA) and above the laser threshold (e.g. I=25 mA) for the temperature range between T=20°C and T=40°C in 5°C steps. Display the respective spectra (if possible in a diagram) and discuss the observed changes. For the spectra above the threshold, also show the dependence of the wavelength on the temperature (diagram) and determine $d\lambda/dT$.
- Use the microscope objective to take a picture of the near field with the camera. Start with low currents (below the laser threshold) and carefully increase the current until laser operation begins. **Always attenuate the light appropriately using gray filters to prevent the camera image from being overexposed!** After taking the near-field image, bring the additional far-field lens into the beam path (see Fig. 7) and take the far-field image.

  Bring a millimeter grid into the beam path in the near field and image it on the CCD by exposing it with an additional light source (e.g. cell phone flashlight) together with the near field of the laser. Estimate the width of the near field using the known grid (d_{square} = 1 mm) and then calculate the width of the laser facet. To do this, the distance between the intermediate image plane and the microscope objective must be measured and be included in the calculation together with the focal length of the microscope objective of f = 6 mm. Describe the resulting far field qualitatively and explain any differences from the near field.

When discussing the results in the test protocol, the following should be addressed in particular in each of the points to be measured:

- What results do I expect according to theory and what results do I get in reality?
- How can any deviations between theory and reality be explained (error analysis)?

4 Literature

[1] Kneubühl, Sigrist, Laser, Teubner
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